

A negative-control outcome calibrates transport bias only when it shares the primary outcome’s bias exactly

A simulation study answering catalog problem IDN-09

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Abstract

Background. Negative-control outcomes detect and calibrate confounding within one data source. Catalog problem IDN-09 asks whether they transfer to transport bias in indirect comparisons, where the bias comes from population differences between studies.

Methods. Unanchored MAIC with an unmeasured covariate V shifted by 0.5 SD between populations or not; a negative-control outcome with no treatment effect whose dependence on V was none, half, equal to or twice the primary outcome’s; and V either modifying A’s effect or not: 16 scenarios, 2000 replicates each. A control test and a calibrated contrast (primary minus control) were compared with naive MAIC.

Results. The control test held its size without a shift (0.039 to 0.056) and flagged the shift in 0.043 to 0.834 of analyses depending on the control’s dependence on V . Calibration removed the bias only when the control’s dependence equaled the primary’s and V did not modify the effect (bias 0.001 against 0.229 naive). With modification it left 0.170; with a control twice as dependent it overcorrected to -0.183, and with modification it happened to cancel (-0.003).

Conclusion. For transport a negative control must share the primary outcome’s bias exactly, including any modification of the treatment effect, which an outcome with no treatment effect cannot do. Use a control signal to detect population differences, not to correct for them.

1 The problem

A negative control (1) shares the confounding pathway of the primary outcome within one population. Across studies the bias is a transport gap: an unmeasured covariate distributed differently between populations. It biases an unanchored MAIC (2) through its prognostic effect on the outcome and through any modification of the treatment effect. A control outcome with no treatment effect can share only the first, and only if its prognostic dependence matches.

2 Design

Registered protocol: `protocol.md`; ADAMP structure (3). Individual data on 500 patients of A; the target publishes B’s (500) proportions for the primary outcome Y and the control N and the mean of a measured covariate. Y : $\text{logit} = -1 + 0.5x + 0.5V$, with B’s effect -0.4 and A’s effect βV ; N : $\text{logit} = -1 + 0.5x + g_N V$. MAIC balances x . The calibrated contrast subtracts the control contrast and adds its variance.

3 Results

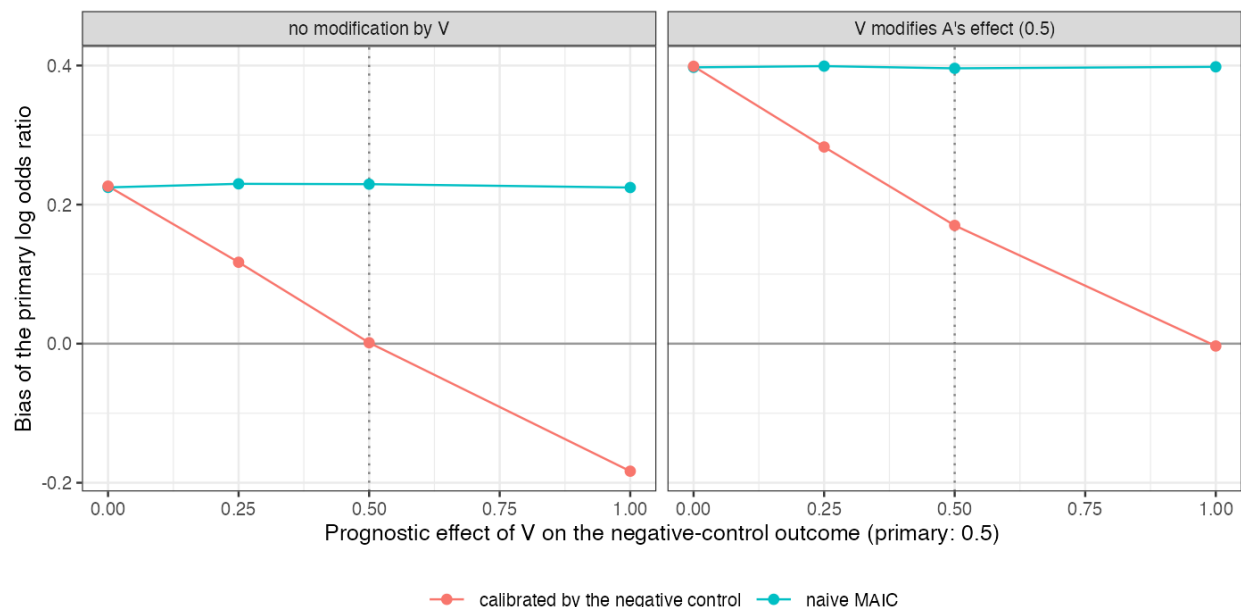


Figure 1: Bias of the naive and calibrated contrasts with a population shift, by the control’s dependence on the unmeasured covariate. The dotted line marks a control matching the primary.

Calibration moved the estimate by the control’s bias whatever the primary’s was (Figure 1). The naive bias came from two sources, prognosis and modification; the control could carry only the first, in proportion to its own dependence on V . The control test’s power rose with that dependence, so the controls that are easiest to detect with are the ones most likely to overcorrect.

4 What this does not answer

One control outcome rather than a bank with empirical calibration; unanchored comparisons only; how often suitable control outcomes are reported in appraisal submissions was not surveyed. Peer review has not been done.

References

1. Marc Lipsitch, Eric Tchetgen Tchetgen, Ted Cohen. Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*. 2010;21(3):383–8. doi:[10.1097/EDE.0b013e3181d61eeb](https://doi.org/10.1097/EDE.0b013e3181d61eeb)
2. James E. Signorovitch, Eric Q. Wu, Andrew P. Yu, Charles M. Gerrits, Evan Kantor, Yanjun Bao, Shiraz R. Gupta, Parvez M. Mulani. Comparative effectiveness without head-to-head trials: A method for matching-adjusted indirect comparisons applied to psoriasis treatment with adalimumab or etanercept. *Pharmacoeconomics*. 2010;28(10):935–45. doi:[10.2165/11538370-000000000-00000](https://doi.org/10.2165/11538370-000000000-00000)
3. Tim P. Morris, Ian R. White, Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074–102. doi:[10.1002/sim.8086](https://doi.org/10.1002/sim.8086)